Class 6
Physics
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1. UNITS & DIMENSIONS

WORKSHEET – 1

LEVEL – I

1. **Answer:** A. Physical quantity
Explanation: By definition, any property of a material, object, or phenomenon that can be quantified through measurement is called a **physical quantity**.

2. **Answer:** B. Mass
Explanation: Mass can be measured using a balance. Weakness, sadness, and affection cannot be measured using standard physical units.

3. **Answer:** A. Fundamental quantities
Explanation: Fundamental (base) quantities are independent physical quantities that cannot be expressed in terms of other physical quantities.

4. **Answer:** B. Derived quantities
Explanation: Derived quantities are obtained by multiplying and/or dividing fundamental quantities.

5. **Answer:** A. Set of fundamental quantities
Explanation: A set of fundamental quantities consists of mutually independent quantities.

6. **Answer:** B. Derived quantities
Explanation:

- Area = Length × Breadth
- Volume = Length^3
- Density = Mass/Volume

Hence, all are derived quantities.

7. **Answer:** A. 2
Explanation: The two supplementary physical quantities are **plane angle** and **solid angle**.

8. **Answer:** A. Fundamental quantities
Explanation: Length, mass, and time are among the seven SI fundamental quantities.

9. **Answer:** A. Area
Explanation: Area is obtained by multiplying two lengths.

10. **Answer:** A. Fundamental quantities
Explanation: Amount of substance (mole) and luminous intensity (candela) are SI fundamental quantities.

LEVEL – II

1. **Answer:** A. Physical quantity
Explanation: A measurable quantity is called a physical quantity.

2. **Answer:** C. {Mass, Length, Time}
Explanation: Mass, length, and time are all fundamental quantities, whereas force and energy are derived quantities.

3. **Answer:** B. Length
Explanation: Length is one of the seven SI base quantities.

4. **Answer:** B. Amount of substance
Explanation: Amount of substance is an SI base quantity, while weight, density, and area are derived quantities.

5. **Answer:** D. Temperature
Explanation: Thermodynamic temperature is one of the seven SI base quantities.

6. **Answer:** C. Plane angle
Explanation: Plane angle and solid angle are the two supplementary quantities.

7. **Answer:** C. Any number of fundamental physical quantities
Explanation: A derived quantity may depend on one or more fundamental quantities.

8. **Answer:** C. Force
Explanation: Force is derived from mass, length, and time.

9. a) Strength of electric current
b) Volume
c) Speed
d) Amount of substance
Answer: B. b & c are correct
Explanation: Volume and speed are derived quantities, whereas electric current and amount of substance are fundamental quantities.

10. a) Luminous intensity
b) Plane angle
c) Area
d) Solid angle
Answer: D. b & d only
Explanation: Plane angle and solid angle are supplementary quantities.

11. **Answer:** A. All derived quantities are obtained by multiplying or dividing the fundamental quantities.
Explanation: Derived quantities are formed by multiplication and/or division of fundamental quantities.

LEVEL – III

Multi-Correct Choice Type

1. **Answer:** A, B, C, D
Explanation: Length, mass, time, and speed are all measurable physical quantities.

2. **Answer:** C. Kindness, D. Pain
Explanation: Kindness and pain cannot be measured using standard physical units.

3. **Answer:** A. Length, B. Mass, C. Time
Explanation: Length, mass, and time are fundamental quantities, whereas density is a derived quantity.

Assertion & Reason

1. **Answer:** C. Statement-I is correct and Statement-II is incorrect.
Explanation: Derived quantities depend on fundamental quantities. Plane angle and solid angle are supplementary quantities, not derived quantities.

Comprehension Type

1. **Answer:** B. Time and Length *(or D. Length and Time, if both options are given)*
Explanation: Distance is a measure of length. Therefore, speed depends on length and time.

2. **Answer:** C. Mass, Length & Time
Explanation: Since acceleration = Length/Time², force depends on mass, length, and time.

3. **Answer:** D. Length
Explanation: Volume = Length × Breadth × Height = Length³. Hence, it is derived only from the fundamental quantity length.

Matrix Match Type

Column-I	**Column-II**
a) Thermodynamic temperature	q) Fundamental quantity
b) Density	r) Derived quantity
c) Solid angle	p) Supplementary quantity

Answer: a → q, b → r, c → p

Integer Type

1. The number of fundamental physical quantities = 7

2. The number of supplementary physical quantities = 2

Worksheet – 2

Level – I

1. **Answer: A**
Explanation: By definition, the established standard used to measure any physical quantity is called its **standard unit**.

2. **Answer: B**
Explanation: Any physical quantity is expressed as the product of its **numerical value (N)** and its **standard unit (U)**.

3. **Answer: B**
Explanation: Since the magnitude of a physical quantity remains constant ($N \times U$ = constant), the numerical value is inversely proportional to the size of the unit: $N \propto \frac{1}{U}$

4. **Answer: A**
Explanation: The complete magnitude of a physical quantity is given by the product of its numerical value and unit: Physical Quantity = $N \times U$

5. **Answer: A**
Explanation: Changing the unit system does not change the actual magnitude of a physical quantity. Only its numerical value changes.

6. **Answer: A**
Explanation: The numerical value indicates how many times the chosen standard unit is contained in the measured quantity.

7. **Answer: D**
Explanation: An ideal standard unit should be **invariable**, **consistent**, and **reproducible** so that measurements remain accurate and uniform everywhere.

8. **Answer: B**
Explanation: A standard unit is the universally accepted reference used for measuring a physical quantity.

Level – II

1. Answer: B

Explanation: Since the magnitude of a physical quantity remains constant,

$$N_1U_1 = N_2U_2$$

Therefore,

$$\frac{N_1}{N_2} = \frac{U_2}{U_1}$$

2. Answer: B

Explanation: Changing the unit system does not alter the actual length of the stick. Therefore,

$$AB = CD$$

3. Answer: A & D

Explanation: Since the magnitude remains constant,

$$PR = QS$$

If unit $R$is smaller than unit $S(R < S)$, then the corresponding numerical value must be larger: $P > Q$

4. Answer: B

Explanation: Given, $PQ = RS$

Therefore, $\frac{Q}{S} = \frac{R}{P}$

If

$$\frac{P}{R} = 100,$$

then

$$\frac{R}{P} = \frac{1}{100}$$

Hence,

$$\frac{Q}{S} = \frac{1}{100}$$

5. Answer: C

Explanation: Fundamental quantities are independent base quantities that cannot be expressed in terms of any other physical quantities.

6. Answer: B

Explanation: A derived quantity is obtained by multiplying or dividing one or more fundamental quantities.

7. Answer: A

Explanation: Since the numerical value **1000** is greater than **1**, and

$$N \propto \frac{1}{U},$$

the unit **kg/m³** must be smaller than the unit **g/cm³**.

8. Answer: D

Explanation: An ideal system of units should be **uniform**, **reproducible**, **invariable**, and **easily accessible**.

9. Answer: B

Explanation: Changing the unit system does not change the actual mass of the boat. Hence, $PQ = RS$

10. Answer: C

Explanation: Since $P \propto \frac{1}{V}$,

it follows that:

- $V \propto \frac{1}{P}$
- $PV =$ constant
- Therefore, $P_1V_1 = P_2V_2$

Multi Correct Choice Type

11. Answer: A, B, C

Explanation: Since the magnitude of a physical quantity remains constant:

- $NU =$ constant
- $N \propto \frac{1}{U}$
- $N_1U_1 = N_2U_2$

All three statements are correct.

12. Answer: A

Explanation: Statement II states the fundamental relationship,

$$N \propto \frac{1}{U},$$

which correctly explains why a larger numerical value corresponds to a smaller unit.

Integer Type

13. Answer: 9

Explanation: In **9 joule**, **joule** is the standard unit and **9** is the numerical value.

14. Answer: 5

Explanation: Using

$$\frac{U_1}{U_2} = \frac{N_2}{N_1},$$

and given

$$\frac{N_1}{N_2} = \frac{1}{5},$$

we get

$$\frac{N_2}{N_1} = 5.$$

Hence,

$$\frac{U_1}{U_2} = 5.$$

Multi Correct Choice Type (Repeated)

15. Answer: A, B, C, D

Explanation: An ideal standard unit should possess all of the following characteristics:

- Consistency
- Reproducibility
- Invariability
- Easy availability

Hence, all four options are correct.

Matrix Match Type

16. Answer: a → r, b → p, c → q, d → s, t

a → r: Physical quantity is given by $P = N \times U$

b → p: Numerical value varies inversely with the unit: $N \propto \frac{1}{U}$

c → q: Conversion between two systems is given by $N_1U_1 = N_2U_2$
d → s, t: If
$\frac{N_1}{N_2} = \frac{2}{5}$,
then
$\frac{U_1}{U_2} = \frac{5}{2}$
and
$\frac{U_2}{U_1} = \frac{2}{5}$.

Comprehension Type

17. Answer: C

Explanation: A physical quantity must include both a numerical value and a unit. A number alone or a unit alone has no physical meaning.

18. Answer: B

Explanation: A physical quantity is expressed as:

$$\boxed{\text{Physical Quantity} = \text{Numerical Value} \times \text{Standard Unit}}$$

19. Answer: D

Explanation: The magnitude of a physical quantity is the product of its numerical value and its standard unit: $\boxed{\text{Physical Quantity} = N \times U}$

Worksheet – 3

LEVEL – I

Single Answer Choice Type

1. Answer: B. 2

Explanation: The SI system includes exactly two supplementary quantities: **plane angle** (measured in **radians**) and **solid angle** (measured in **steradians**).

2. Answer: C. Mass

Explanation: Mass is one of the **seven fundamental (base) physical quantities** in the SI system. Plane angle and solid angle are supplementary quantities, while density is a derived quantity.

3. Answer: C. Force

Explanation: Force is a derived quantity because it is obtained from fundamental quantities using the relation: Force = Mass × Acceleration
Mass, time, and temperature are fundamental quantities.

4. Answer: D. All of these

Explanation: Force, work, and power are all derived quantities because they are obtained by combining fundamental quantities.

5. Answer: D. SI

Explanation: The **International System of Units (SI)** is the universally accepted standard system of measurement used worldwide.

6. Answer: D. 7

Explanation: The seven fundamental physical quantities are:

- Length
- Mass
- Time
- Electric current
- Thermodynamic temperature

- Amount of substance
- Luminous intensity

7. Answer: C. Newton
Explanation: Newton (N) is the derived SI unit of force.

$$1\,\mathrm{N} = 1\,\mathrm{kg\,m/s^2}$$

Ampere, candela, and kelvin are fundamental SI units.

8. Answer: B. Candela
Explanation: Candela (cd) is the SI base unit of luminous intensity.

9. Answer: A. Radian
Explanation: Plane angle is measured in **radians (rad)**, while solid angle is measured in **steradians (sr)**.

10. Answer: A. N
Explanation: Force is measured in **newton**, whose symbol is **N** (capital letter because it is named after Sir Isaac Newton).

LEVEL – II

(Single Answer Choice Type)

11. Answer: C. Unit
Explanation: Every physical quantity is measured by comparing it with a standard known as a **unit**.

12. Answer: A. Foot
Explanation: In the Foot–Pound–Second (FPS) system, length is measured in **foot**.

13. Answer: B. Pound
Explanation: In the FPS system, the unit of mass is **pound**.

14. Answer: C. Centimetre
Explanation: In the Centimetre–Gram–Second (CGS) system, the unit of length is **centimetre (cm)**.

15. Answer: A. Kilogram
Explanation: In the Metre–Kilogram–Second (MKS) system, mass is measured in **kilograms (kg)**.

16. Answer: A. SI
Explanation: SI (International System of Units) is the globally accepted standard system of units.

17. Answer: D. Metre
Explanation: In the MKS system, the unit of length is **metre (m)**.

18. Answer: A. Gram
Explanation: In the CGS system, mass is measured in **grams (g)**.

19. Answer: B. Second
Explanation: Time is measured in **seconds (s)** in both the CGS and SI systems.

20. Answer: C. cm
Explanation: The standard symbol for centimetre is **cm**, written in lowercase.

21. Answer: D. m/s
Explanation: Metre per second (m/s) is a derived unit of speed because it combines the units of length and time.

LEVEL – III

Multiple Correct Choice Type

22. Answer: A, B, D

Explanation:

- **A:** Correct – There are **7 fundamental quantities** in SI.
- **B:** Correct – MKS has three basic quantities: length, mass, and time.
- **D:** Correct – The base units (m, kg, s) are common to both MKS and SI.
- **C:** Incorrect.

23. Answer: A, B, C

Explanation:

- Metre (m), kilogram (kg), and second (s) are fundamental SI units.
- m/s is a derived unit.

24. Answer: A, B, C

Explanation:

- Candela (cd), mole (mol), and kelvin (K) are SI base units.
- kg/m^3 is the derived unit of density.

25. Answer: C, D

Explanation:

- Ampere (A) and kelvin (K) are SI base units.
- Newton (N) and joule (J) are derived units.

26. Answer: A. Both Statements I and II are correct.

Explanation: Both statements are correct and represent basic definitions of the SI system.

27. Answer: D. Statement I is incorrect and Statement II is correct.

Explanation: Derived units depend on combinations of fundamental units and are therefore **not independent**.

Matrix Match Type

28. Answer: a → s, b → t, c → r, d → p, q

a → s: m^3 is the unit of volume (derived unit).

b → t: Kilogram is a fundamental unit.

c → r: Area is a derived quantity.

d → p, q: Height and thickness are both measurements of length.

29. Answer: a → r, s, b → q, c → t, d → p

Length → centimetre/metre

Time → second

Plane angle → radian

Electric current → ampere

Integer Type

30. Answer: 2

Explanation: Volume is measured in cubic metres. $m \times m \times m = m^3$

Thus, metre is multiplied by itself **2 additional times**.

31. Answer: 1

Explanation: Area is measured in square metres.

$$m \times m = m^2$$

Thus, metre is multiplied by itself **1 additional time**.

Worksheet – 4

LEVEL – I

Single Answer Choice Type

1. **Correct Answer:** D. All the above
Explanation: The SI base unit for length is the metre (m). Micron (10^{-6} m), Fermi (10^{-15} m), and Light year (9.46×10^{15} m) are commonly used units of length, but none of them are SI units.

2. **Correct Answer:** A. k
Explanation: The SI prefix for kilo (10^3) is represented by the lowercase **k** (e.g., kg, km). The uppercase **K** is the symbol for kelvin.

3. **Correct Answer:** B. milli
Explanation: The SI prefix **milli** represents one-thousandth of a unit, i.e., 10^{-3}.

4. **Correct Answer:** C. 10^{-6}
Explanation: The SI prefix **micro** (μ) represents one-millionth of the base unit, i.e., 10^{-6}.

5. **Correct Answer:** C. millimetre
Explanation: Kilometre, hectometre, and decametre are multiples of metre, whereas millimetre (10^{-3} m) is a sub-multiple.

6. **Correct Answer:** C. nano
Explanation: Nano (10^{-9}) represents a value smaller than the base unit and is therefore a sub-multiple.

7. **Correct Answer:** C. deca
Explanation: Deca represents 10^1, making it a multiple of the base unit.

8. **Correct Answer:** A. tera
Explanation: Tera represents 10^{12}, which is greater than giga (10^9), mega (10^6), and kilo (10^3).

9. **Correct Answer:** C. exa
Explanation: Exa represents 10^{18}, making it the largest among the given prefixes.

10. **Correct Answer:** B. sub-multiple
Explanation: Quantities smaller than the standard unit are expressed using sub-multiples.

LEVEL – II

Single Answer Choice Type

11. **Correct Answer:** A. Light year
Explanation: A light-year is the distance travelled by light in vacuum during one year.

12. **Correct Answer:** A. 10^{-15}m
Explanation: One fermi (or one femtometre) is equal to 10^{-15}metre.

13. **Correct Answer:** D. 1
Explanation: Since 1 μm = 10^{-6}m, the thickness is **1 micron**.

14. **Correct Answer:** A. centi
Explanation: The SI prefix **centi** represents 10^{-2}.

15. **Correct Answer:** D. 10^3
Explanation: One tonne is equal to **1000 kg** or 10^3kg.

16. **Correct Answer:** B. amu
Explanation: The mass of atoms is conveniently expressed in **atomic mass unit (amu)**.

17. **Correct Answer:** C. 400×10^{-9}m
Explanation: Nano (n) represents 10^{-9}. Therefore,
$$400 \text{ nm} = 400 \times 10^{-9} \text{ m}$$

18. **Correct Answer:** C. femto
Explanation: The SI prefix **femto** represents 10^{-15}.

19. **Correct Answer:** D. μm
Explanation: Plant cells are typically measured in micrometres (μm).

20. **Correct Answer:** B. 1000
Explanation: Since 1 mm = 10^{-3} m,
1 m = 1000 mm

LEVEL – III

Multi Correct Choice Type

1. **Correct Answer:** A, B, C, D (All are correct)
Explanation: Micrometre (10^{-6} m), nanometre (10^{-9} m), decimetre (10^{-1} m), and centimetre (10^{-2} m) are all sub-multiples of metre.

2. **Correct Answer:** A, B, C, D (All are correct)
Explanation:
A. 1 g = 10^{-3} kg
B. 1 cm = 10^{-2} m
C. 1 m = 10^{3} mm
D. 1 g = 10^{6} μg
All the above statements are correct.

Statement Type

3. **Correct Answer:** A. Both Statement-I and Statement-II are correct.
Explanation:
Statement-I is correct because 1 km = 1000 m.
Statement-II is also correct because 1 parsec ≈ 3.26 light-years.

4. **Correct Answer:** A. Both Statement-I and Statement-II are correct.
Explanation: Both are standard SI prefix values.

Comprehension Type

5. **Correct Answer:** A. decametre
Explanation: A decametre equals 10 metres, making it a multiple of metre.

6. **Correct Answer:** B. milligram
Explanation: Milligram is a smaller unit of mass than kilogram.

7. **Correct Answer:** B. millimetre
Explanation: Millimetre (10^{-3} m) is a sub-multiple of metre.

8. **Correct Answer:** D. 10^{12}m
Explanation: The prefix **tera** represents 10^{12}.

Integer Type

9. **Correct Answer:** 2
Explanation: Since **hecto = 100 =** 10^{2},
1 hm = 10^{2} m, Therefore, $n = 2$.

10. Correct Answer: 1
Explanation: There are 3600 seconds in 1 hour.

Matrix Match Type

11. Match the following
Correct Matching: a → q, s, b → t, c → p, d → r

Explanation:

- Shake is a non-standard unit of time equal to 10^{-8}s.
- Pound is an imperial unit of mass equal to approximately 0.453 kg.
- Micron is another name for micrometre, equal to 10^{-6}m.
- Light-year is a unit of distance equal to approximately 9.46×10^{15}m.

Worksheet – 5

LEVEL – I

1. Answer: B. K
The SI base unit of thermodynamic temperature is the kelvin, whose symbol is K (uppercase, without the degree symbol). The symbol θ represents the dimension of temperature, not its unit.

2. Answer: A. A
The SI base unit of electric current is the ampere, represented by the symbol A. The symbol I denotes electric current in equations, while C is the unit of electric charge (coulomb).

3. Answer: B. cd
The SI base unit of luminous intensity is the candela, represented by the symbol cd (both letters in lowercase).

4. Answer: C. mol
The SI base unit of amount of substance is the mole, whose symbol is mol.

5. Answer: B. Dimensional formula
A dimensional formula expresses a physical quantity in terms of the powers of the fundamental quantities such as mass (M), length (L), and time (T). For example, the dimensional formula of force is $[MLT^{-2}]$.

6. Answer: A. Dimensions
The dimensions of a physical quantity are the exponents of the fundamental quantities present in its dimensional formula.

7. Answer: D. Change in velocity / Time
Acceleration is defined as the rate of change of velocity with time.

$$\boxed{a = \frac{\Delta v}{t}}$$

8. Answer: D. Mass × Velocity
Momentum is the product of the mass of an object and its velocity.

$$\boxed{p = mv}$$

9. Answer: A. Mass × Acceleration
According to Newton's Second Law of Motion, $\boxed{F = ma}$

10. Answer: D. Force × Time
Impulse is the product of force and the time for which it acts.

$$\boxed{\text{Impulse} = F \times t}$$

LEVEL – II

11. Answer: C. [$M^0L^2T^0$]

Area = Length × Breadth

$= L \times L = L^2$

Therefore, $\boxed{[M^0L^2T^0]}$

12. Answer: B. [$M^0L^3T^0$]

Volume = Length × Breadth × Height

$= L \times L \times L = L^3$

Hence, $\boxed{[M^0L^3T^0]}$

13. Answer: B. [M^0LT^{-1}]

$$\text{Velocity} = \frac{\text{Displacement}}{\text{Time}} = \frac{L}{T} = LT^{-1}$$

Therefore, $\boxed{[M^0LT^{-1}]}$

14. Answer: C. [$ML^{-3}T^0$]

$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{L^3} = ML^{-3}$

Hence, $\boxed{[ML^{-3}T^0]}$

15. Answer: D. [M^0LT^{-2}]

$$\text{Acceleration} = \frac{\text{Velocity}}{\text{Time}} = \frac{LT^{-1}}{T} = LT^{-2}$$

Therefore, $\boxed{[M^0LT^{-2}]}$

16. Answer: D. Length

The dimensional formula of length is $[L] = [M^0L^1T^0]$

The exponent of time is zero.

17. Answer: C. –3

$$\text{Power} = \frac{\text{Work}}{\text{Time}} = \frac{[ML^2T^{-2}]}{T} = [ML^2T^{-3}]$$

Hence, the exponent of time is –3.

18. Answer: B. [MLT^{-2}]

$F = ma = M \times LT^{-2}$

Therefore, $\boxed{[MLT^{-2}]}$

19. Answer: D. All of the above

- Area = [L^2]
- Volume = [L^3]
- Speed = [LT^{-1}]

All have zero power of mass.

20. Answer: B. 2

Power = $[ML^2T^{-3}]$

The exponent of length is 2.

LEVEL – III

Multi-Correct Choice Type

21. Answer: A. Speed and B. Velocity

Both have the dimensional formula $\boxed{[LT^{-1}]}$

22. Answer: A. Acceleration, B. Force and D. Work

Quantity	Dimensional Formula	Time Dimension
Acceleration	$[LT^{-2}]$	–2
Force	$[MLT^{-2}]$	–2
Work	$[ML^2T^{-2}]$	–2
Velocity	$[LT^{-1}]$	–1

Hence, Acceleration, Force and Work have the same time dimension.

23. Answer: A. Both statements are true.

Volume has the dimensional formula $[L^3]$

Therefore, the exponent of length is 3. Statement-II correctly defines dimensions.

24. Answer: A. 1

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{[MLT^{-2}]}{[L^2]} = [ML^{-1}T^{-2}]$$

The exponent of mass is 1.

25. Answer: B. –1

Pressure has the dimensional formula

$$[ML^{-1}T^{-2}]$$

The exponent of length is –1.

26. Answer: D. –2

Pressure has the dimensional formula $[ML^{-1}T^{-2}]$

Hence, the exponent of time is –2.

27. Answer: C. Derived quantity

If force, length and time are taken as fundamental quantities, then mass is obtained from $F = ma$

Therefore, mass becomes a derived quantity.

28. Answer: A. Fundamental quantity

India follows the SI system, in which mass is one of the seven fundamental (base) quantities.

29. Answer: D. newton

The SI (MKS) unit of force is the newton (N).

Matrix Match Type

30. Match Column-I with Column-II

Answer: a → t, b → s, c → r, d → p, e → q

These are the standard dimensional symbols of the corresponding SI base quantities. (Note: The dimension symbol for amount of substance is [N], not [mol]. mol is the SI unit symbol.)

Integer Type

31. Answer: 1

Work = Force × Distance = $[MLT^{-2}] \times [L] = [ML^2T^{-2}]$

Hence, the exponent of mass is 1.

Worksheet – 6

Level I

1. Answer: B. Length
By definition, **length** is the measurement of the distance or spatial separation between two distinct points.

2. Answer: B. metre
The **metre (m)** is the SI base unit of length.

3. Answer: A. Same physical quantities
Unit conversion is possible only between units representing the **same physical quantity** (e.g., cm to m, kg to g). Different physical quantities cannot be converted into one another.

4. Answer: C. Mass
Mass is the amount of matter present in a body. It is independent of gravity and remains constant.

5. Answer: D. kilogram
The **kilogram (kg)** is the SI base unit of mass.

6. Answer: B. 60 minute
One hour consists of **60 minutes**.

7. Answer: D. centimetre
The length of a book is conveniently measured in **centimetres (cm)**.

8. Answer: A. Physical balance
A **physical balance (or analytical balance)** is used for accurately measuring small masses such as gold.

9. Answer: A. Wrist watch
A **wrist watch** is an instrument used to measure time.

10. Answer: C. 100
1 m = 100 cm

Level – II

11. Answer: B. 1000
1 m = 1000 mm

12. Answer: B. 1000
The prefix **kilo-** represents 10^3.
1 km = 1000 m

13. Answer: A. 5

$$500 \text{ cm} = \frac{500}{100} \text{ m} = 5 \text{ m}$$

14. Answer: 3×10^7 nm
$3 \text{ cm} = 3 \times 10^{-2}$ m
Since,
$1 \text{ nm} = 10^{-9}$ m
$3 \text{ cm} = 3 \times 10^{-2} \times 10^9 = 3 \times 10^7$ nm

15. Answer: A. 40×10^2 cm
$40 \text{ m} = 40 \times 100 \text{ cm} = 4000 \text{ cm} = 40 \times 10^2$ cm
The remaining options are incorrect.

16. Answer: A. 1000 g

1 kg = 1000 g

17. Answer: C. 5×10^8

$500 \text{ kg} = 500 \times 10^3 \text{ g}$

$= 500 \times 10^6 \text{ mg} = 5 \times 10^8 \text{ mg}$

18. Answer: C. 10^9

$1 \text{ kg} = 10^3 \text{ g}$

$1 \text{ g} = 10^6 \mu\text{g}$

$1 \text{ kg} = 10^3 \times 10^6 = 10^9 \mu\text{g}$

19. Answer: B. 100 kg

1 quintal = 100 kg

20. Answer: A. $\frac{1}{1000}$ kg

$1 \text{ g} = 10^{-3} \text{ kg} = \frac{1}{1000} \text{ kg}$

21. Answer: C. 1000 kg

1 metric ton = 1000 kg

22. Answer: D. 10

$1 \text{ metric ton} = \frac{1000}{100} = 10 \text{ quintals}$

23. Answer: C. 10^{-6}

The prefix **micro (µ)** represents 10^{-6}.

$1 \mu\text{g} = 10^{-6} \text{ g}$

24. Answer: B. 10^{-5}

$1 \text{ shake} = 10^{-8} \text{ s}$

Since,

$1 \text{ ms} = 10^{-3} \text{ s}$

$1 \text{ shake} = \frac{10^{-8}}{10^{-3}} = 10^{-5} \text{ ms}$

25. Answer: B. 10^{-6}

$1 \text{ ng} = 10^{-9} \text{ g}$

$1 \text{ mg} = 10^{-3} \text{ g}$

$1 \text{ ng} = \frac{10^{-9}}{10^{-3}} = 10^{-6} \text{ mg}$

26. Answer: A, B and C

$1 \mu\text{g} = 10^{-6} \text{ g}$

Verification:

- **A:** $10^{-3} \text{ mg} = 10^{-6} \text{ g}$
- **B:** $10^3 \text{ ng} = 10^{-6} \text{ g}$
- **C:** 10^{-6} g

27. Answer: A, B and C

$1 \text{ Mm} = 10^6 \text{ m}$

Verification:

- **A:** $10^3 \text{ km} = 10^6 \text{ m}$
- **B:** 10^6 m

- **C:** 10^{15} nm = 10^{6} m

28. Answer: C. Statement I is correct and Statement II is incorrect

Statement I: True

A physical quantity is given by $Q = n \times u$

If the unit changes, the numerical value also changes so that the quantity remains constant.

Statement II: False

The numerical value is **inversely proportional** to the size of the unit.

For example, 1 m = 100 cm

A smaller unit gives a larger numerical value.

29. Answer: −6

1 ng = 10^{-6} mg

Hence, $x = -6$

30. Answer: −12

1 ng = 10^{-9} g

1 g = 10^{-3} kg

Therefore, 1 ng = $10^{-9} \times 10^{-3} = 10^{-12}$ kg

Hence, $x = -12$

31. Answer: A. 10^{4}

1 cm = 10 mm

1 ns = 10^{-3} μs

Therefore,

$$\frac{1 \text{ cm}}{1 \text{ ns}} = \frac{10 \text{ mm}}{10^{-3}\ \mu\text{s}} = 10^{4} \text{ mm}/\mu\text{s}$$

32. Answer: B or C

1 mg = 10^{3} μg

1 km = 10^{6} mm

Therefore,

$$\frac{1 \text{ mg}}{1 \text{ km}} = \frac{10^{3}\ \mu\text{g}}{10^{6} \text{ mm}} = 10^{-3}\ \mu\text{g/mm}$$

33. Answer: C. 10^{9}

1 μs = 10^{-3} ms

1 nm = 10^{-12} km

Therefore,

$$\frac{1\ \mu\text{s}}{1 \text{ nm}} = \frac{10^{-3} \text{ ms}}{10^{-12} \text{ km}} = 10^{9} \text{ ms/km}$$

34. Answer: C. 0.2×10^{-3}

20 cm = 0.2 m

0.2 m = 0.2×10^{-3} km = 2×10^{-4} km

35. Answer: C. 5×10^{-6}

5 mg = 5×10^{-3} g

= $5 \times 10^{-3} \times 10^{-3}$ kg = 5×10^{-6} kg

36. Matrix Match Type

Answer: a → r, b → s, c → r, d → s

Worksheet – 7

Level – I

1. **Answer:** C
Area is defined as the measurement of the total two-dimensional surface space occupied by an object.

2. **Answer:** C
In the International System of Units (SI), the fundamental unit for length is the metre (m). Consequently, the derived unit for area is the square metre (m^2).

3. **Answer:** A
Since 1 m = 100 cm = 10^2 cm, squaring both sides gives:
$$(1\text{ m})^2 = (10^2\text{ cm})^2 = 10^4\text{ cm}^2$$

4. **Answer:** B
Since 1 cm = 10^{-2} m, squaring both sides gives:
$$(1\text{ cm})^2 = (10^{-2}\text{ m})^2 = 10^{-4}\text{ m}^2$$

5. **Answer:** D
The area of a square is given by: Area = side × side
Substituting the given value: 1 m × 1 m = 1 m^2
Thus, the area is **1 square metre**.

6: **Answer:** B
100 m × 100 m = 10,000 m^2
Since 10,000 m^2 = 1 hectare,
the required answer is **1 hectare**.

7. **Answer:** C
The SI unit of force is **Newton (N)** and the CGS unit is **dyne**.
Since 1 N = 10^5 dyne,
the ratio of SI unit to CGS unit is $10^5 : 1$.

8. **Answer:** B
1 kg = 10^6 mg
and
1 m^3 = 10^6 cm^3.
Therefore,
$$\frac{1\text{ kg}}{1\text{ m}^3} = \frac{10^6\text{ mg}}{10^6\text{ cm}^3} = 1\text{ mg cm}^{-3}.$$

9. **Answer:** C
Since 1 m = 100 cm = 10^2 cm,
cubing both sides gives $(1\text{ m})^3 = (10^2\text{ cm})^3 = 10^6\text{ cm}^3$.

10. **Answer:** D
Since 1 cm = 10^{-2} m,
cubing both sides gives $(1\text{ cm})^3 = (10^{-2}\text{ m})^3 = 10^{-6}\text{ m}^3$.

LEVEL – II

1. **Answer:** A
Since 1 cm = 10^{-2} m,
$40\text{ cm}^3 = 40 \times (10^{-2}\text{ m})^3 = 40 \times 10^{-6}\text{ m}^3 = 4 \times 10^{-5}\text{ m}^3$.

2. **Answer:** C

Since 1 m = 10^3 mm,
cubing both sides gives 1 m^3 = 10^9 mm^3.
Therefore, 3 m^3 = 3×10^9 mm^3.

3. **Answer:** B
1 hectare = 10^4 m^2
and 1 m^2 = 10^6 mm^2.
Hence, 1 hectare = $10^4 \times 10^6 = 10^{10}$ mm^2.

4. **Answer:** B
$1\mu m = 10^{-6}$ m = 10^{-9} km
Therefore, $1\mu m^2 = (10^{-9}\ km)^2 = 10^{-18}\ km^2$.
Hence, $x = -18$.

5. **Answer:** A
1 mm = $10^3\mu m$
Squaring both sides,1 mm^2 = $(10^3)^2 = 10^6\mu m^2$.

6. **Answer:** D
1 km = 10^5 cm
Therefore, 1 km^2 = $(10^5)^2 = 10^{10}$ cm^2.

7. **Answer:** C
1 cm = 10 mm
Squaring both sides,
1 cm^2 = 10^2 mm^2 = 100 mm^2.

8. **Answer:** D
1 m = 10^3 mm
Squaring both sides, 1 m^2 = 10^6 mm^2.

9. **Answer:** B
1 hectare = 10^4 m^2 = 10^8 cm^2.
Therefore,
$10^{x+2} = 10^8$
$x + 2 = 8$
$x = 6$.

10. **Answer:** B
Since
1 litre = 10^{-3} m^3,
1000 litres = 1000×10^{-3} m^3 = 1 m^3.

LEVEL – III

Multi Correct Choice Type

11. **Answer:** A, D
1 cm^2 = 10^{-4} m^2
which is also equal to
$\frac{1}{10000}$ m^2.

Hence, both **Option A** and **Option D** are correct.

12. **Answer:** A, B, C
$1\ m^2 = 10^4\ cm^2$
$1\ m^2 = 10^6\ mm^2$
$1\ m^2 = 10^{-6}\ km^2$.
Therefore, **Options A, B and C** are correct.

13. **Answer:** A, B, C
$1\ km^2 = 10^6\ m^2$
$1\ km^2 = 10^{10}\ cm^2$
$1\ km^2 = 10^{12}\ mm^2$.
Hence, **Options A, B and C** are correct.

14. **Answer:** A, B, C
10 hectares $= 10 \times 10^4 = 10^5\ m^2$
which is equal to
$10^9\ cm^2$,
$10^{11}\ mm^2$,
and
$10^{-1}\ km^2$.
Therefore, **Options A, B and C** are correct.

Reasoning Type

15. **Answer:** A
Statement I is true because $(10^3\ m)^2 = 10^6\ m^2$.
Statement II is also true because the **hectare** is a standard unit used for measuring large areas of land.

16. **Answer:** A
Convert breadth into metres: $2000\ cm = 20\ m$.
Area $= 500\ m \times 20\ m = 10{,}000\ m^2 = 10^4\ m^2$.

17. **Answer:** B
Volume $= 20 \times 18 \times 15 = 5400\ cm^3$.

18. **Answer:** A
Density is given by

$$\text{Density} = \frac{\text{Mass}}{\text{Volume}} = \frac{232}{20} = 11.6\ g/cm^3.$$

19. **Matrix Match Type**
Answer: a → q, b → r, c → s, d → p
a) Area of square → **q)** side × side
b) Area of rectangle → **r)** length × breadth
c) Volume of cube → **s)** side × side × side
d) Volume of cuboid → **p)** length × breadth × height

Integer Type

20. **Answer:** 4
$1\ m^2 = 10{,}000\ cm^2 = 10^4\ cm^2$.
Hence, $n = 4$.

21. **Answer:** 1
By definition, 1 hectare $= 10{,}000\ m^2$.
Therefore, $10000\ m^2 = 1$ hectare.

Worksheet – 8

LEVEL – I

1. Answer: C

Standard vernier callipers have a least count of **0.1 mm**, which can also be written as $\frac{1}{10}$ mm.

2. The principle of vernier is

Answer: B

The standard design principle of a vernier scale is: nV.S.D = $(n-1)$M.S.D where **V.S.D** is Vernier Scale Division and **M.S.D** is Main Scale Division.

3. Answer: A

The smallest division marked directly on the main scale is called the **Main Scale Division (M.S.D)**.

4. Answer: B

The smallest measurement that an instrument can measure accurately is called its **least count**.

5. Answer: A

The least count is given by L.C = 1 M.S.D – 1 V.S.D

6. Answer: A

Alternatively,

$$\text{L.C} = \frac{\text{Pitch}}{\text{Number of Vernier Scale Divisions}}$$

where Pitch = value of one Main Scale Division.

7. Answer: C

An ordinary metric scale has a least count of **1 mm (0.1 cm)**.

8. Answer: B

The commonly used laboratory vernier callipers have a least count of **0.1 mm (0.01 cm)**.

9. Answer: C

The external jaws are used to measure **external dimensions**, such as the external diameter or thickness of an object.

10. Answer: A

The **internal jaws** are used to measure internal diameters and widths.

LEVEL – II

11. Answer: B

$$\text{Pitch} = \frac{1 \text{ cm}}{20} = 0.05 \text{ cm}$$

12. Answer: A

The Main Scale Reading (M.S.R) is the reading just before the zero of the vernier scale.

Hence, $\boxed{2.6 \text{ cm}}$

13. Answer: B

$$\text{Pitch} = \frac{1 \text{ cm}}{10} = 0.1 \text{ cm}$$

14. Answer: B

Vernier callipers are also called **slide callipers** because the vernier scale slides over the main scale.

15. Answer: A

Length = M.S.R + (V.S.C × L.C)
where
M.S.R = Main Scale Reading
V.S.C = Vernier Scale Coincidence
L.C = Least Count

16. Answer: B

The metallic strip (depth rod) is used for measuring the **depth of vessels or holes**.

17. Answer: A

The difference between **1 M.S.D** and **1 V.S.D** is called the **least count**.

18. Answer: C

A vernier calliper consists of:

- Main Scale
- Vernier Scale

19. Answer: B

M.S.R = 4 mm = 0.4 cm

L.C = $\frac{1 \text{ mm}}{10}$ = 0.1 mm = 0.01 cm

Length = $0.4 + (6 \times 0.01) = 0.46$ cm

20. Answer: D

$d = a + bn$

This follows from

Measurement = M.S.R + (V.S.C × L.C)

Reasoning Type

21. Answer: D

Statement-I is incorrect because the difference between **1 M.S.D** and **1 V.S.D** is called the **least count**, not the pitch. Statement-II is correct.

Matrix Match Type

22. Column-I vs Column-II

Answer: a → r, t, b → p, c → s, d → q

(a) Principle: nV.S.D = $(n-1)$M.S.D
or specifically, 10V.S.D = 9M.S.D
(b) L.C = 1 M.S.D – 1 V.S.D
(c) Internal jaws measure internal dimensions.

- (d) Metallic strip measures the depth of a vessel.

LEVEL – III

Multi Correct Choice Type

23. Answer: A, D

$\frac{1}{10}$ mm = 0.1 mm

Since

1 mm = 0.1 cm

0.1 mm = 0.01 cm

Hence both options are correct.

24. Answer: A, B

$$\text{Pitch} = \frac{1 \text{ cm}}{10} = 0.1 \text{ cm}$$

$$\text{L.C} = \frac{0.1}{10} = 0.01 \text{ cm}$$

Reasoning Type

25. Statement-I: nV.S.D = $(n-1)$M.S.D.

Statement-II: The smallest value of the length which can be read directly from a main scale is pitch.

Answer: A

Both statements are scientifically correct and independent facts regarding the construction and measurement principles of a vernier calliper.

Comprehension Type

26. Answer: A (OR D)

The standard least count of a vernier calliper is **0.1 mm**, which is numerically the same as **0.10 mm**. Hence both options are equivalent.

27. Answer: C

The metallic strip (depth rod) is used to measure the **depth of vessels, containers, or holes**.

28. Answer: B

Internal jaws are used for measuring **internal diameters and internal widths**.

Matrix Match Type

29. Column-I vs Column-II

Answer: a → p, t, b → r, c → q, d → s

(a) Both formulas represent methods of calculating the least count.

(b) The expression is equivalent to M.S.R + (V.S.C × L.C)

since M.S.R = Number of Main Scale Divisions × Pitch

(c) Positive zero error is added as V.S.C × L.C

(d) Negative zero error is calculated as $(n$ – Coinciding Division) × L.C

Integer Type

30. Answer: 1

$$\text{Pitch} = \frac{1 \text{ cm}}{10} = 0.1 \text{ cm} = 1 \times 10^{-1} \text{ cm}$$

31. Answer: 9

From nV.S.D = $(n-1)$M.S.D

For $n = 10$, 10V.S.D = 9M.S.D

32. Answer: 1

Pitch = 0.1 cm = 1 mm

$$\text{L.C} = \frac{1 \text{ mm}}{10} = 0.1 \text{ mm}$$

$0.1 \text{ mm} = 10^{-4} \text{ m} = 1 \times 10^{-4} \text{ m}$

Hence, the required integer is $\boxed{1}$

2. Vectors

Worksheet – 1

Level – I

1. A	2. C	3. C	4. C	5. A	6. D
7. B	8. B	9. A	10. A	11. A	12. C

Level – II

1. A	2. B	3. C	4. D	5. C	6. D
7. C	8. D	9. C	10. D	11. B	12. B
13. A	14. D	15. D	16. B	17. A, B, C	18. B, D
19. C, D	20. A	21. C	22. D	23. A-Q,R,T B-P,S C-P,S D-P,S	
24. A-S, B-S, C-Q, D-Q		25. C	26. A, B	27. B	28. A
29. C	30. 150^0				

3. KINEMATICS

Level – I

1. A	2. C	3. B	4. A	5. B	6. D
7. D	8. B	9. A			

Level -II

1.A	2.B	3. C	4. D	5. A	6. A
7. C	8. B	9. 1-D	10. 2-D	11. A	12. A
13. A	14. B	15. B	16. D	17. D	18. D

Level – III

19. A, B	20. A	22. B	23. a – p; b – q; c – r; d – s
24. A, B, C, D	25. C	26. A	

SOLUTIONS

Level – I

1. In motion with respect to the wall
2. Inner wall of train
3. Motion
4. Rest
5. Relative
6. all of the above
7. Rectilinear Motion
8. Rotatory
9. 1 Dimensional

Level -II

1. Rectilinear
2. Translatory
3. Circular
4. Rectilinear Motion
5. Vibrational Motion
6. Linear Motion
7. Rotatory
8. Periodic
9. Circular Motion
10. Periodic
11. Non - Periodic
12. Periodic
13. Non - Periodic
14. Rectilinear Motion 17) Rotatary motion & translatory motion
15. Rotational Motion Translatory Motion
16. Motion 22) Rest
17. Rest
18. $a-p, b-q, c-r, d-s$ 24) 1,2,3

Level – III

19. Both A & B are correct
20. Both (A) and (B) true and (R) is correct explanation of (A)

WORKSHEET – 2

Level – I

1.A	2. A	3. A	4. A	5. B	6. B
7. B	8. C	9. A	10. B		

Level – II

1.A	2. D	3. A	4. B	5. B	6. D
7. D	8. A	9. 5	10. D		

Level -III

11. A	12. C	13. C	14. C	15. D	16. B
17. A	18. A	19. B	20. C	21. D	
22. a-q; b-p; c-r; d-s		23. A 24. A, B		25. B, C	26. B
27. B	28. B	29. a-r; b-s; c-t; d-q, r,			

SOLUTIONS

Level -I

1. Distance
2. Scalar
3. Magnitude
4. cm
5. M
6. Displacement
7. Vector
8. Both (1) & (2)
9. cm
10. m

Level – II

1. Magnitude of displacement = 0
2. It must be Zero
3. May or may not be zero
4. Zero
5. $2\pi r$
6. $4 + 3 + 12 = 19$ m
7. May be ≤ 1
8. A body may travel some distance without having displacement
9. $10 - 5 = 5$ km
10. $2 + 2 = 4$

Level – III

11. $8 + 6 = 14$ m
12. $AC = \sqrt{(AB)^2 + (BC)^2}$
    $AC = \sqrt{(8)^2 + (6)^2} = \sqrt{64 + 36}$
    $= \sqrt{100} = 10$ m
18. $55 - 50 = 5m$
19. $5 + 5 = 10m$
20. $5 + 5 + 5 = 15m$
21. $5 + 5 + 5 + 5 = 20$ m
22. $a - q, b - p, c - r, d - s$
23. Both 1&2 are correct
24. Displacement is zero
    Distance cover is $2\pi R$
25. Displacement of car is 75 m$100 - 25 = 75$ m
    Distance is $100 + 25 = 125$ m

4. KINEMATICS – II

Worksheet – 1

Level – I

1. C	2. B	3. B	4. A	5. A	6. A
7. A	8. C	9. C	10. A		

Level – II

1. **Answer: B**

$$375 \times \frac{18}{56} = \frac{54 \text{ km}}{\text{hr}}$$

2. **Answer: B**

$$S = \frac{d}{t} = \frac{120 \text{ m}}{2\text{sec}} = 60 \text{ m/s}$$

3. **Answer: c**

$$d = s \times t = 70 \times 3 = 210 \text{ km}$$

4. **Answer: A**

Zero

5. **Answer: A**

$$0.52.5 \times \frac{18}{} = 9 \text{ km}$$

$$d = s \times t = 9 \times 1.5 = 13.5 \text{ km}$$

6. **Answer: D**

$$t = \frac{d}{s} = \frac{20}{10} = 2\text{hr}$$

7. Answer: **A**

$$s = \frac{d}{t} = \frac{\Pi r}{2} = — = \Pi\text{m/s}$$

8. **Answer**: **A**

$$\text{speed} = \frac{2\pi r}{t}$$

9. **3m/s**

$$t = 2 \times 60 = 120\text{sec}$$

$$s = \frac{d}{t} = \frac{360}{120} = 3 \text{ m/s}$$

10. **Answer**: **B**

$$d = 2 + 2 + 2 + 2 = 8m \text{ t} = 4\text{sec}$$

$$s = \frac{d}{t} = \frac{8}{4} = 2 \text{ m/s}$$

11. **Answer: C**

$$d_1 = s \times t = 1 \times 60 = 60 \text{ m}$$

$$d_2 = s \times t = 3 \times 60 = 180 \text{ m}$$

$$\text{Total distance} = d_1 + d_2$$

$$= 60 + 80 = 240 \text{ m}$$

12. **Answer: D**

$$s = {}^{4}72 \times \frac{5}{18} = 20 \text{ m/s}$$

$$t = \frac{d}{s} = \frac{100}{20} = 5\text{sec}$$

13. Answer: A

$$t_1 = \frac{d}{s} = \frac{60}{120} = \frac{1}{2}\text{hr}$$

$$t_2 = \frac{d}{s} = \frac{60}{80} = \frac{3}{4}\text{hr}$$

$$T = T_1 + T_2$$

$$= \frac{1}{2} + \frac{3}{4} = \frac{10}{8} = 1.25 \times 60 = 75 \text{ min}$$

14. Answer: A

$d = s \times t = 10 \times 15 = 150$ m

15. Answer: A

$d = 8 + 6 = 14$ m

$t = 14$ s

$$s = \frac{d}{t} = \frac{14}{14} = 1 \text{ m/s}$$

Level – III

16. Answer: D

$$s = \frac{d}{t} = \frac{30}{30} = 1 \text{ m/s}$$

17. Answer: A

$$t = \frac{d}{s} = \frac{15}{120} = 0.125 \text{ h}$$

18. Answer: D

$d = s \times t = 15 \times 45 = 675$ m

19. Answer:

a-q; b-r; c-p; d-s

20. Answer:

A, B, C, D

21. Answer: 6M/S

$$T_1 = 6 \text{ m/s} = \frac{60}{10}$$

22. Answer:

A, B

23. Answer: A

$s = d/t = 10/10 = 1$ cm/s

24. Answer: **C**

$t = d/s = 100/5 = 20$sec

25. Answer: **A**

$d = s \times t = 20 \times 20 = 400$ m

26. Answer: C

$$\text{average speed} = \frac{(160-100)\text{km}}{1\text{hr}}$$

$= 60$ km/hr

27. Answer: B

$$\text{average speed} = \frac{(220-160)\text{km}}{2\text{hr}}$$

$= \frac{60}{2} = 30$ km/hr

28. Answer: B

average speed $= \frac{(220-100)\text{km}}{3\text{hr}}$

$= \frac{120}{3} = 40$ km/hr

29. Answer: A, B

average speed $= \frac{500\text{ m}}{25\text{ s}} = 20$ m/s

$= 20 \times \frac{18}{5} = 72$ km/hr

30. Answer:

a-r; b-s; c-q; d-p

Worksheet – 2

Level – I

1. C	2. B	3. A	4. B	5. D	6. C
7. B	8. A	9. B	10		

Level – II

1. Answer: C

$30 \times \frac{18}{5} = 6 \times 18 = 108$ km/h

2. Answer: A

$72 \times \frac{5}{18} = 4 \times 5 = 20$ m/s

3. Answer: D

4. Answer: A

$v = \frac{d}{t} = \frac{100}{2} = 50$ m/s

5. Answer: C

$v = \frac{d}{t} = \frac{15}{2} = 7.5$ m/s

6. Answer: B

$t = 5 \times 60 = 300$sec

7. Answer: B

8. Answer: 1sec

$v = \frac{d}{t} \Rightarrow t = \frac{d}{v} = \frac{10}{10} = 1$sec

9. Answer: B

$v = \frac{d}{t} \Rightarrow t = \frac{2r}{t} = \frac{2(2)}{2} = 2$ m/s

10. Answer: A

11. Answer: A

$v = \frac{d}{t} = \frac{6}{6} = 1$ cm/s

12. Answer: B

13. Answer: 4m/s

$V = \frac{2r}{t} = \frac{2 \times 120}{60} = 4$ m/s

14. Answer: B

$r = 10$ cm

t = 1 minutes $\quad v = \frac{2r}{t} = \frac{2\times10}{1 \text{ minute}} = 20$ cm/min

15. Answer: B

18kmph $\rightarrow$ 5 m/s

$\xrightarrow[V_1]{}$ E

W $\xleftarrow{V_2}$

72kmph $\rightarrow$ 25 m/s

$\Delta V = V_2 - V_1$

$= 20 - (-5) = 25$ m/s

$t = \frac{s}{\Delta v} = \frac{500}{25} = 20$ s

Level – III

16. Answer: A

17. Answer: C

S = 500 kmt = 1hr

$V = \frac{d}{t} = \frac{500}{1} \Rightarrow 500$ km/hr

18. Answer: A

$V = \frac{2r}{t}$

19. Answer: a-q; b-s; c-p; d-r

20. Answer: A, B, C

SI: m/s

CGS: cm/s km/hr.

21. Answer: A, B

22. Answer: A

Moving with uniform velocity. Instantaneous speed will be same.

23. Answer: D

5 m/s

24. Answer: A

At the end also 5 m/s.

H is maintaining 5 m/s for 10 minutes.

25. Answer: A, B, C

26. Answer: a-p,q; b-p,q; c-r,s; d-r,s

27. Answer: D

$\text{Averagevelocity} = \frac{\text{total displacement}}{\text{total time}}$

$= \frac{s_1 + s_2}{t_1 + t_2} = \frac{2 \text{ s}}{s\left[\frac{1}{v_1} + \frac{1}{v_2}\right]}$

$= \frac{2}{\frac{v_1 + v_2}{v_1 v_2}} = \frac{2v_1 v_2}{v_1 + v_2}$

28. Answer: D

40 m along OX

30 m perpendicular to OX

So, hyp $= \sqrt{40^2 + 30^2}$

$\Rightarrow \sqrt{2500} = 50$ m

t = 10 s and $v = \frac{50}{10} = 5$ m/s

29. Answer: B

$$\text{Averagevelocity} = \frac{\text{total displacement}}{\text{total time}}$$

Worksheet – 3

Level – I

1. A	2. C	3. A	4. C	5. B	6. A
7. D	8. A	9. B	10. B		

Level – II

1. Answer: A

$\Delta v = v_2 - v_1 = 20 - 10 = 10$ m/s

2. Answer: B

$\Delta v = v_2 - v_1 \Rightarrow v_1 = v_2 - \Delta v$
$= 40 - 20 = 20$ m/s

3. Answer: D

$\Delta v = v_2 - v_1 \Rightarrow v_2 = \Delta v + v_1$
$= 100 + 30 = 130$ m/s

4. Answer: A

5. Answer: C

6. Answer: C

$\Delta v = v_2 - v_1 = 20 - 0 = 20$ m/s

7. Answer: D

$\Delta v = v_2 - v_1 = -v_2 - v_1$
$= -5 - 10 = -15$ m/s

8. Answer: D

$\Delta v = v_2 - v_1$
$= v_2 - v_1 = 20 - 20 = -40$ m/s

9. Answer: D

$\Delta v = v_2 - v_1$

$v_1 = 72\text{kmph} \Rightarrow 72 \times \frac{5}{18} = 20$ m/s

$v_2 = 90\text{kmph} \Rightarrow 90 \times \frac{5}{18} = 25$ m/s

$\Delta v = -25 - 20 = -45$ m/s

10. Answer: C

$\Delta v = v_2 - v_1 = -100 - 50 = -150$ m/s

11. Answer: D

$\Delta V = v_2 - v_1 = -100 - 120 = -220$

12. Answer: C

$\Delta v = v_2 - v_1$
$= -27.77 - 20 = -47.77$ m/s

13. Answer: A

Level – III

14. Answer: A, C

15. Answer: A, B, C

16. Answer: D

$\Delta v = v_2 - V_1$

$= -75 - 60 = -135$ m/s

17. Answer: A

$\Delta v = v_2 - v_1 = 75 - 60 = 15$ m/s

18. Answer: D

$\Delta V = v_2 - v_1 = -105 - 80$

$\Delta V = -185$kmph

19. Answer: $a - r; b - p; c - q; d - s$

20. Answer: C

$|\Delta v| = |v_2 - v_1| = |2 - 5| = |-3| = 3$kmph

21. Answer: A

$v_1 = 36\text{kmph} = 36 \times \frac{5}{18} = 10$ m/s

22. Answer: B

$v_2 = 108\text{kmph} = 108 \times \frac{5}{18} = 30$ m/s $= 3000$ cm/s

23. Answer: A

$\Delta v = v_2 - v_1 = 30 - 10 = 20$ m/s

24. Answer: 5

$\Delta v = v_2 - v_1$

$v_2 = 18 \times \frac{5}{18} = 5$ m/s

$\Delta v = 5 - 0 = 5$ m/s

25. Answer: a-r;b-s;c-p;d-t

a) $200 \times \frac{5}{18} = \frac{500}{9} = 55.5$ m/s

b) $250 \times \frac{5}{18} = 69.44$ m/s $= 6944$ cm/s

c) $= 55.5 \times 100$ cm/s $= 55.50$ cm/s

d) $= 6944 \times 100$ m/s $= 6944$ cm/s

26. Answer: B, C

$\Delta v = v_2 - v_1$

$= -288 - 270 = -588$kmph

$= 588 \times \frac{5}{18} = 163.33$ m/s

27. Answer: B

$V_1 = 90\text{kmph} = 90 \times \frac{5}{18} = 25$ m/s

28. Answer: C

$v_2 = 72\text{kmph} = 72 \times \frac{5}{18}$

$v_2 = 20$ m/s $= 2000$ cm

29. Answer: B

$|72 - 90| = |-18| = 18$kmph

Worksheet – 4

Level – I

1. C 2. C 3. B 4. A 5. B 6. B
7. 2 8. D 9. A 10. D

Level – II

1. Answer: C

2. Answer: A

3. Answer: C

$$a = \frac{v_2 - v_1}{t} = \frac{6 - 12}{3} = \frac{-6}{3} = -2 \text{ m/s}^2$$

4. Answer: A

$a = -a \Rightarrow -10 \text{ m/s}^2$

5. Answer: D

6. Answer: B

7. Answer: B

$$v_1 = 108\text{kmph} = 108 \times \frac{5}{18} = 30 \text{ m/s}$$

8. Answer: A

9. Answer: B

10. Answer: B

11. Answer: D

$$v_0 = 0, v_2 = 54\text{kmph} = 54 \times \frac{5}{18} = 15 \text{ m/s}$$

$$t = 5\text{sec}$$

$$a = \frac{v_2 - v_1}{t} = \frac{15 - 0}{5} = 3 \text{ m/s}^2$$

12. Answer: B

$a = \frac{v_2 - v_1}{t}, v_1 = 20 \text{ m/s}, v_2 = 20 \text{ m/s}$ $t = 10\text{sec}$

$$a = \frac{20 - 20}{10} = 0 \text{ m/s}^2$$

13. Answer: B

$$a = \frac{v_2 - v_1}{t} = \frac{70 - 40}{5 \times \frac{1}{60}}$$

$$= \frac{30 \times 60}{5} = 360 \text{ km/h}^2$$

14. Answer: D

$$\frac{a_1 t_1 + a_2 t_2}{t_1 + t^2}$$

15. Answer: 10

$a = \frac{v_2 - v_1}{t}, v_1 = 10 \text{ m/s}$

$$v_2 = 108\text{kmph} = 108 \times \frac{5}{18} = 30 \text{ m/s}$$

$t = 2\text{sec}$

$$a = \frac{30 - 10}{2} = \frac{20}{2} = 10 \text{ m/s}^2$$

16. Answer: A, B

17. Answer: A, B, C, D

18. Answer: D

$a = \frac{v_2 - v_1}{t}, v_1 = 10 \text{ m/s}, v_2 = 0, t = 5\text{sec}$

$= \frac{0 - 10}{5} = \frac{-10}{5} = -2 \text{ m/s}^2$

$-a = 2 \text{ m/s}^2$

19. Answer: C

$a = \frac{v_2 - v_1}{t}, v_1 = 15 \text{ m/s},$

$v_2 = 5 \text{ m/s}, t = 2\text{sec}$

$= \frac{5 - 15}{2} = \frac{-10}{2} = -5 \text{ m/s}^2$

$-a = 5 \text{ m/s}^2$

20. Answer: C

$\text{Acceleration} = -5 \text{ m/s}^2$

21. Answer: 0.1

$u = \frac{v - u}{t} = \frac{4 - 2}{20} = \frac{2}{20} = 0.1 \text{ m/s}^2$

22. Answer: 0.1 a-p,q;b-r,t;c-r,s;d-q,s

a) $a = \frac{v-u}{t} = \frac{5-10}{2} = \frac{-5}{2} - = 2.5 \text{ m/s}^2$

b) $a = \frac{v-u}{T} = \frac{25-15}{5} = \frac{10}{5} = 2 \text{ m/s}^2$

c) $a = \frac{v-u}{t} = \frac{5-0}{5} = 1 \text{ m/s}^2$

d) $a = \frac{v-u}{t} = \frac{0-10}{10} = 1 \text{ m/s}^2$

23. Answer: A, B, C, D

24. Answer: B

$a = \frac{v_2 - v_1}{t}, v_1 = 72 \times \frac{5}{18} = 20 \text{ m/s}$

$v_2 = 0, t = 25$

$a = \frac{0 - 20}{25} = \frac{-20}{25} = -0.8 \text{ m/s}^2$

25. Answer: D

26. Answer: C

27. Answer: A

28. Answer: A, D

29. Answer: a-p;b-q;c-r;d-s

30. Answer: 5

5. TYPES OF FORCE

WORKSHEET – 1

LEVEL – I

1. A	2. A	3. A	4. B	5. B	6. C
7. D	8. C	9. D	10. A		

2. The distance covered by coin will be loss compared to passanger and train, as is acceleration but coin is moving with same speed in horizontal directio.
3. The average force exerted on the chinaware is small and chances of its breaking reduce.
4. In uniform circular motion there is a change in direction of the velocity and acceleration is constant.

LEVEL – II

1. D	2. C	3. D	4. B	5. A	6. C
7. A	8. D	9. B	10. A	11. B	12. D
13. D	14. C	15. C	16. C		

2. Collision in between bat and ball, momentum takes places loss of energy is max because of light bat.
3. When the tanker's brakes are applied, the velocity of the tanker becomes less but there is no external force acting directly on the water. The K.E. from the water didn't go away when the brakes are applied. It is similar to slamming the brakes in your car. Your body moves forward & you need the seat belt to stop you.
4. $1\ \text{N} = 1\ \text{kg} \cdot \text{m/s}^2$

   $= 1\ \text{kg} \cdot \text{wt}$

   $= 9.8\ \text{N}$ (or) $10\ \text{N}$

   $1\ \text{N} = \frac{1}{10}\ \text{kg}$

   $= \frac{1}{10} \times 10^3\ \text{g}$

   $= 10^2\ \text{g}$ (or) $100\ \text{g}$

LEVEL – III

17. B, C	18. B	19. B	20. D
21. A → S, B →R, C → P, D → Q		22. 102GM	23. A, D
24. A, B, D	25. A	26. B	27. C

27. $F = ma$

$a \rightarrow$ constant

$u = 0$

$S = ut + \frac{1}{2}att^2$

$S = \frac{1}{2}att^2$

$S \alpha t^2$

WORKSHEET – 2

LEVEL – I

1. A	2. C	3. B	4. A	5. B	6) 4
7. C	8. B	9. C	10. B		

LEVEL – II

1. C	2. D	3. D	4. C	5. 4	6. D
7. A	8. B	9. C	10. C	11. D	12. A
13. D	14. D	15. C	16. D		

LEVEL – II

17. A, B, C, D	18. B	19. A	20. C
21. $A \rightarrow Q, B \rightarrow R, C \rightarrow S, D \rightarrow P$		22.2.5 M/S^2	
23. A, B, D	24. C	25. B	26. A
27. A, B, D	28. C	29. B	30. D

SOLUTIONS

6. Mass = 20 kg

$\bar{a} = 2\ m/s^2$

$F = 20 \times 2 = 40$ N.

7. $F = 15$ N, $m = 15$ kg $\cdot$ $u = 0, t = 8sec, v = ?$

$F = m\left(\frac{v-u}{t}\right)$

$15 = 15\left(\frac{v-0}{8}\right)$

$v = 8$ m/s

9. $20 = 10\left(\frac{2v-v}{8}\right)$

$v = 16$ m/s

10. $F = \frac{1}{10} \times \left(\frac{0-20}{0.01}\right) = 200$ N

11. $F = 80\left(\frac{25-5}{2}\right) = 800$ N

12. $F = 4kgf = 4 \times 9.8 = 39.2$

$A = \frac{F}{m} = \frac{39.2}{20} = 1.96 m/s^2$

14. $F_1 = m_1 a_1$

$m_1 = \frac{F}{a} \Rightarrow \frac{5}{8}$

$F_2 = m_2 a_2$

$m_2 = \frac{5}{24}.$

$m_1 \rightarrow m_2 \rightarrow f = 5N.$

$5 = (m_1 + m_2)a$

$a = \frac{5}{\frac{5}{8} + \frac{5}{24}}$

6

$a = \frac{5}{\frac{15+5}{24}} = \frac{5}{\frac{20}{24}} = \frac{5 \times 24}{20}$

$a = 6 \text{ m/s}^2$

15. $F = m\left(\frac{v-u}{t}\right)$

$F = \frac{150}{1000}\left(\frac{20+12}{0.02}\right)$

$F = 240N$

16. Initial Speed $u = 36$ km/h

$= 36 \times \frac{5}{18} = 10$ m/s

Final Speed = 0 m/s.

t = 4 s

$m = 400 \text{ kg}, m^1 = 65 \text{ kg}$

Total mass = $m + m^1$

$= 400 + 65 = 465$ kg.

Using NLM Ist Law.

v = u + at

$a = \frac{v-u}{t} = \frac{0-10}{4} = -2.5 \text{ m/s}^2$

[∵ - ve sign indicates that the velocity of the three wheeler is decreasing with time).

$F = m \times a$

$= 465 \times (2.5)$

$F = -1162.5$ N

-ve sign indicates that the force is against the direction of motion.

18. $F = 200\left(\frac{0-19.6}{10}\right)$

$F = 392N$

19. $F = m\left(\frac{v-u}{t}\right)$

$= 196$ N

20. $F_1 = \frac{392}{9.8} = 40$kgf

$F_2 = \frac{196}{9.8} = 20kgf$

22. $250 = 100(a)$

$a = \frac{5}{2} = 2.5$ m/s^2

24. $a = \frac{F}{m_1+m_2+m_3} = \frac{6.5}{3.1+2.4+1.2}$

$= 0.97$ m/s^2

25. $F_{31} = m_3a = (1.2)(0.97)$

$= (1.164)\left(\frac{1}{1}\right) = 1.164$ N

Rounding add to two significant the force exerted by the recorder on third car would be 1.2 N.

26. $(2.4 + 1.2)(0.97)$

$= 3.492 \times \frac{1}{1} = 3.492 = 3.5$ N

28. $F = m\left(\frac{v-u}{t}\right)$

29. $8000 = 2(a)$

$a = 4000$ m/s^2

30. $F = ma = 8000N$

WORKSHEET - 3

LEVEL – I

1. D 2. C 3. D 4. A 5. B 6. B
7. D 8. A 9. B 10. A

LEVEL – II

1. C 2. C 3. D 4. A 5. B 6. A
7. C 8. B 9. D 10. B 11. A 12. B
13. A 14. C 15. B 16. D 17. D 18. C
19. B

LEVEL – III

20. b, d 21. a → r, b →s, c→p, d →q 22. 6.67→10→11
23. a, c 24. b 25. b 26. D 27. a, d 28. *a→s, b→t, c→ p, d→q*

10. $F = \frac{Gm_1m_2}{d^2}$ $F' = \frac{Gm_1m_2}{\frac{d^2}{4}}$

$\frac{F}{F^1} = \frac{1}{4}$

$F^1 = 4F$

11. $F = \frac{Gm_1m_2}{d^2}$

$F = \frac{Gm_1\frac{m_2}{2}}{d^2}$ 2

from (1) & (2) $d_1 = \frac{d}{\sqrt{2}}$

13. $F = \frac{6.67\times10^{-11}\times1\times2}{(50\times10^{-2})^2}$

$F = 5.3 \times 10^{-10}$ N.

14. When d increases, F decreases

$F\alpha\frac{m_1\ m_2}{d^2}$

15. $F = \frac{6.67\times10^{-11}\times1\times1}{(10^{-2})^2}$

$F = 6.67 \times 10^{-7}$ N.

16. Gravitational feru acts towards the centre of the object.

17. $F_1 = \frac{Gm_1m_2}{r^2}$

$F_2 = \frac{Gm_1m_2}{(3r)^2}$

$F_2 = \frac{F_1}{9}$

19. $F = \frac{Gm^2}{r^2}$

$F^1 = \frac{G^{m^2}/2}{r_1^2}$

$F = F^1$

$r_1^2 = \frac{r^2}{2} \Rightarrow r_1 = \frac{r}{\sqrt{2}}$.

25. $F = \frac{Gm_1m_2}{r^2}$

$= 6.67 \times 10^{-14} \times \frac{1\times2}{1\times1}$

100 cm = 1 m

26. $F = \frac{1\times4\times G}{4}$

$F = G$

27. $F = \frac{mv-mu}{t}$

$v = 0 \Rightarrow$ body comes to rest after force in applied.

$F = \frac{-mu}{t}$

WORKSHEET – 4

LEVEL – I

1. D	2. C	3. D	4. C	5. B	6. A
7. A	8. C	9. D	10. B		

LEVEL – II

1. C	2. C	3. D	4. C	5. B	6. A
7. B	8. A	9. C	10. B	11. A	12. D
13. B	14. A	15. C	16. 3	17. B	18. B

LEVEL – III

19. A, B	20. a-r; b-s; c-p; d-q	21. 0.25 g/cm3		22. A, B, D
23. B, D	24. a-s,t; b-r; c-p; d-q	25. C	26. D	27. A

SOLUTIONS

14. In CGS System density of Kerosene is 0.8gm cm^{-3}

In SI System $0.8 \times 1000 \frac{kg}{m^3}$

$= 800\ kg/m^3$

15. Object wt in air = 36 g

Buoyant force = $V_i e_L g$

$= 8 \times 1 \times g = 8g$

wt in air = $36g - 8g = 28g$.

18. $d = \frac{m}{v}$

$v = \frac{m}{d} = \frac{900}{3} = 300\ cm^3$

23. $A = 4\ cm^2$

$\therefore s_s = 1500\ kg/m^3$

$I = 60\ cm$

$\therefore S_w = 1000\ kg/m^3 = 10^3\ kg/m^3$

$v_i = Al$

$\therefore g = 10\ m/s^2$

$v_i = 4 \times 60\ cm^3$

$v_i = 240 \times 10^{-6}\ m^3$

upthrust $(F) = V_i P_L\ g$

$= 240 \times 10^{-6}\ m^3 \times 10^3\ kg/m^{3 \times 10\ m/s^2}$

upthrust $(F) = 2.4$ N

Weight $= V_i P_s g = 240 \times 10^{-6} \times 1500 \times 10$

Weight = 3.6 N

25. $m = 0.5$ kg, $g = 9.8$ m/s^2

(F) Buoyant Force = Weight = mg

$= 0.5 \times 9.8$

$F = 4.9$ N

26. $m = 0.5$ kg

$\rho_L = 10^3$ kg/m^3

$V_i = ?$

density $(\rho) = \frac{\text{mass(m)}}{\text{volume(v)}}$ (or) $\rho = \frac{m}{v}$

$v = \frac{m}{s}$

$= \frac{0.5}{10^3} = 5 \times 10^{-4}$ m^3

27. V of ball $= \frac{4}{3}\pi r^3, r = 11$ cm $= 0.11$ m

$V = 5.58 \times 10^{-3}$ m^3

$m = 0.5$ kg

$\rho = \frac{m}{V}$

$\rho = \frac{0.5}{5.58 \times 10^{-3}} = 90$ kg/m^3

6. NEWTON'S LAWS OF MOTION

WORKSHEET – 1

LEVEL – I

1. C	2. C	3. B	4. C	5. C	6. C
7. A	8. B	9. C	10. C		

LEVEL – II

1. B	2. B	3. D	4. C	5. B	6. B
7. C	8. C	9. C	10. C	11. A	12. 2
13. C	14. C	15. A			

Level – III

16. A-R; B-Q; C-P; D-S	17. A, B, C	18. A	19. D	20. B
21. C	22. B	23. C	24. A	

WORKSHEET – 2

LEVEL – I

1. D	2. A	3. B	4. D	5. C	6. C
7. 2	8. B	9. A	10. C		

LEVEL – II

1. A	2. B	3. A	4. B	5. D	6. D
7. B	8. B	9. C	10. D	11. B	12. A
13. D	14. D	15. C	16. D	17. C	

Level – III

18. A	19. A	20. A	21. A 22. a-t; b-r; c-p; d-	23.D
24. D	25. C	26. C	27. C	28. C

29. a-p,q,r,s; b-p,q,r,s; c-p,r; d-q,s

SOLUTIONS

LEVEL – II

1. $P = MV = 10 \times 10 = 100$ kg m/s
2. $P = MV = 15 \times 5 = 75$ kg m/s
3. $p = mv$
4. $KE = \frac{P^2}{2\ m}$

   $4 = \frac{P^2}{2 \times 2}$

   $P^2 = 16, P = 4NS$
5. $KE = \frac{P^2}{2m} \Rightarrow KE\alpha\frac{1}{m}$
6. $m = 1$ kg, $p = 2$Ns

   $p = mv$

$2 = 1 \times v = v = 2$ m/s

$\mathrm{KE} = \frac{\mathrm{P}^2}{2\ \mathrm{m}} = \frac{4}{2} = 2$ J

7. m = 20gm

$v = 5$ cm/s

P = mv = 100gmcm/s

$p = mv = 20 \times 5 = 100$ g cm/s.

8. momentum

= (mass) × (Initial velocity) = $mu. = p_i$

9. $P_f = mv$

10. $\Delta P = P_f - P_i = mv - mu$

11. $P = mv$

12. m = 20 kg

p = 800 kg m/s

v =?

p = mv

$800 = 20 \times v$

v = 40 m/s

13. $KE = P.E = P$

$KE = \frac{p^2}{2m}$

$P = \frac{P^2}{2m}$

$p = 2m$

m$v = 2$ m

v = 2 m/s

14. $P = mv$

$KE = \frac{p^2}{2m}$

$\frac{KE_1}{KE_2} = \left(\frac{p_1}{p_2}\right)^2 \times \left(\frac{m_2}{m_1}\right)$

$\frac{8}{1} = \left(\frac{p_1}{p_2}\right)^2 \times \left(\frac{m}{2m}\right)$

$16 = \left(\frac{p_1}{p_2}\right)^2$ $(p_1 : p_2) = 4 : 1$

15. $P_1 = P\ P_2 = 2P$

$\frac{\mathrm{KE}_1}{\mathrm{KE}_2} = \left(\frac{\mathrm{P}_1}{\mathrm{P}_2}\right)$

$\frac{\mathrm{KE}}{2\mathrm{KE}} = \left(\frac{\mathrm{P}_1}{\mathrm{P}_2}\right)^2$

$\frac{1}{2} = \left(\frac{P_1}{P_2}\right)^2$

$\frac{1}{2} = \left(\frac{P_1}{P_2}\right)^2$

$\frac{\mathrm{P}_1}{\mathrm{P}_2} = \frac{1}{\sqrt{2}}$

$P_2 = \sqrt{2}P$

16. $\frac{\mathrm{KE}_1}{\mathrm{KE}_2} = \frac{4}{1}$

$P_1 = P_2$
$m_1 = m_2 = ?$
$KE = \frac{p^2}{2m}$
$KE = \alpha \frac{1}{m}$
$\frac{KE_1}{KE_2} = \frac{m_2}{m_1} = \frac{4}{1}$
$\frac{m_1}{m_2} = 1:4$

17. $r_1 = r_2 = 1:2$
$p_1 = p_2 = 1:1$
$KE_1 = KE_2 = ?$
$KE = \frac{p^2}{2\ m}$
$KE \alpha \frac{1}{m}\left[\begin{matrix} m = vp \\ m = \frac{4}{3}\pi r^3 p \end{matrix}\right]$
$\frac{KE_1}{KE_2} = \left(\frac{r_2}{r_1}\right)^3 = \left(\frac{2}{1}\right)^3 = \frac{8}{1}$
$KE_1 : KE_2 = 8:1$

LEVEL – III

18. $M = 200\ g v = 5\ m/s$
$p = 2mv = 0.2 \times 5 = 1 kgm/s$
19. $p_i = mu$
20. $p_f = mv$
21. $\Delta p = mv - mu = m(v - u)$
22. a-t; b-r; c-p; d-s
23. $\frac{p_1}{p_2} = \frac{m_1 v_1}{m_2 v_2} = \frac{v_1}{v_2}\ (\because m_1 = m_2)$
24. $\frac{p_1}{p_2} = \frac{v_1}{v_2} = \frac{v}{3v} = \frac{1}{3}$
25. $m = \frac{p}{v} = \frac{20}{5} = 4\ kg$
26. $p_i = mu = 3 \times 10 = 30\ kg\ m/s$
27. $p_f = mu = 3 \times 20 = 60\ kg\ m/s$
28. $\Delta P = P_f - P_i = 60 - 30 = 30\ kg\ m/s$

WORKSHEET – 3

1. D	2. A	3. A	4. B	5. B	6. C
7. A	8. C	9. A	10. C		

Level – II

1.D	2.C	3. A	4. B	5. D	6. A
7. A	8. B	9. C	10. C	11. B	12. B
13. C	14. A	15. B	16. B	17. B	18. C

LEVEL – III

19. C 20. C 21. B 22. B 23. C 24. B

25. a-s; b-r; c- q; d-p 26.105 27. A 28. A 29. A

30. A, C, D 31. a- p, q; b-r, s; c-p, q, r, s; d-t

SOLUTIONS

Level – I

9. $\Delta p = f \times t = 12 \times 4 = 48$ kg m/s

Level – II

1. $F = Ma \Rightarrow 250 = 100 \times a$
   $a = 2.5\ m/s^2$
2. 1 kgf $= 9.8$ N $\simeq 10$ N
   $20 \times 9.8 = 2 \times a$
   $a = 100\ m/s^2$
3. $F = ma \Rightarrow 250 = 10a$
   $a = 25\ m/s^2$
4. 1 gf $= 980$ dyne
   $F = ma$
   $3 \times 980 = m \times 980 = 3g$
6. $20 \times 9.8 = 19.6 \times m$
   $m = 10$ kg
7. $F = ma$
   $200 = 1\ kg \times a$
   $a = 200\ m/s^2$
10. $F = \frac{m(v-u)}{t}$
    $p_i = m(u) = 5(0) = 0$kgm/s
    $p_f = m(v) = 5(10) = 50$kgm/s
11. $f = ma$
    $f = (200 \times 10^{-3})(10) = 2$ N
    $f = 2 \times 10^5$ dyne
13. $f = m\left(\frac{v-u}{t}\right)$
    $0.6 = \frac{(6-3)m}{2}$
    $m = 0.4$ kg
14. $\frac{F_1}{F_2} = \frac{m_1 a_1}{m_2 a_2} = \frac{(2)(2.5)}{(2)(5)} = \frac{5}{10}$ $(F_1,\ F_2) = (5\ N; 10\ N)$
15. $F = ma$
    $F = (30)(4) = 120$ N
    $F = 120\ N = 120 \times 10^5$ dyne
    $\therefore F = 120 \times 10^5$ dyne
16. $F = ma$
    $F = (20)(20) = 4 \times 10^2$ dyne
    $F = 4 \times 10^2 (10^{-5} N) = 4 \times 10^{-3}$
18. $S = \frac{1}{2}\frac{f}{m}t^2$

$400 = \frac{1}{2}\frac{F}{200}$

$F = 4 \times 10^4$ dyne

$F = 4 \times 10^4 \times 10^{-5}$ N

$F = 4 \times 10^{-1}$ N $= 0.4$ N

Level – III

19. $F = ma$

$g = ma$

$g = \left(\frac{W}{g}\right)a [\because W = mg]$

$w = \frac{g^2}{a}$

20. $a = \frac{f}{m}, a = \frac{v-u}{t}$

21. $F = \frac{\Delta p}{t} = \frac{m \cdot \Delta v}{t} = \frac{(2)4}{6} = \frac{4}{3}$

22. $a = \frac{f}{m}$

23. $\Delta P = m(v - u)$

24. $F = ma$

25. $\Delta P = F \times t = 100 \times 10 = 1000$kgm/s

26. $\Delta p = m \cdot \Delta v$

$10^3 = (20)\Delta v$

$\therefore \Delta v = 50$

29. $a = \frac{F}{m} = \frac{100}{20} = 5$ m/s^2

WORKSHEET – 4

1. A	2. C	3. B	4. B	5. D	6. C
7. C	8. A	9. D			

Level – II

1. C	2. A	3. D	4. C	5. B	6. C
7. C	8. B	9. B	10. C	11. C	12. C
13. C	14. B	15. C	16. B	17. B	18. A

Level – III

19. A, B, C	20. A, B, C, D	21. C	22. a-t; b-q; c-r, d-p,s	23. A
24. D	25. a-q,s; b-p; c-r	26. B	27. B	

WORKSHEET – 5

Level – I

1. A	2. D	3. B	4. A	5. A	6. D
7. D	8. A	9. B	10. D		

Level – II

1. B	2. A	3. A	4. B	5. C	6. C
7. D	8. D	9. B	10. C	11. A	12. B
13. D	14. D	15. A	16. B		

Level – III

17. C, D	18. D	19. D	20. A	21. a-q,p;b-p,q;c-r,t;d-s
22.C	23. A	24. B	25. D	26. A 27. D

Solutions

LEVEL – II

1. $I = F.t$
   $= 10.3$
   $I = 30Ns$
2. $I = F.t$
   $= 200 \times 8$
   $= 1600\text{dyne} - S$
3. $mv - mu = F \times t$
4. $\Delta P = F.t$
   $= (2.5)(0.1)$
   $= 0.25Ns$
5. $\Delta P = F.t$
   $= (30)(0.1)$
   $= 3Ns$
6. $F.t = \Delta P$
   $4 \times 2 = \Delta P$
   $\Delta P = I$
   $I = 8Ns$
7. $F = \frac{\Delta P}{\Delta t}$
   $\Delta P = I$
   $F = \frac{100}{0.2}$
   $F = 50N$
8. $I = F.t$
   $= 200 \times 10^{-5} \times 0.3$
   $= 100 \times 10^{-6}$
   $= 0.0006$
9. $\Delta P = I = F.t$
   $I = 8 \times 0.2$
   $= 1.6Ns$
10. $I = F \times t$
    $= 2.5 \times 0.01$
    $= 0.025$
11. $F = 2\ N$
    $m = 4\ kg$
    $t = 1sec$
    $a = \frac{F}{m}$
    $= \frac{2}{4} = 0.5\ m/s^2$
    $v = u + at$
    $v = 0 + (0.5)(1)$
    $v = 0.5\ m/s$

$m = 3 \times 10^{-3}$ kg
$t = 5$sec
$v = 100$ m/s
$F_1 = \frac{m_1 v_1}{t_1}, F_2 \frac{m_2 v_2}{t_2}$
$F_1 = F_2$
$\frac{m_1 v_1}{t_1} = \frac{m_2 v_2}{t_2}$ $\frac{30 \times 10^{-3} \times 100}{5} = \frac{10 \times v_2}{2 \times 60}$
$v_2 = 7.2$ m/s

14. $\frac{m_1 v_1}{t_1} = \frac{m_2 v_2}{t_2}$
$\frac{(2)(10)}{0.5} = \frac{(10)v_2}{60}$
$v_2 = 240$ m/s

15. $u = 72 \times \frac{5}{18} = 20$ m/s
$V = 0.$
$\Delta P = mv - mu$
$= m(v - u)$
$= 2000(20 - 0)$
$= 4 \times 10^4$Ns

16. $m = 100$ kg
$u = 30$ m/s
$v = 20$ m/s
$\Delta p = m(v - (-u))$
$\Delta p = m(v + u)$
$= 100(20 + 30)$
$= 100 \times 50$
$= 5000$ N

LEVEL – III

17. Conceptual
18. $I = F \times t$
$= 5 \times 10$
$= 50$Ns
19. $I = \Delta P$
$\Delta P = 50$Ns
20. $I = m\Delta v$
$50 = 2\Delta v$
$\Delta v = \frac{50}{2} = 25$ m/s
21. Conceptual
22. $I = F \times t$
$= 200 \times 10^{-5} \times 20$
$= 4000 \times 10^{-5}$
$I = 4 \times 10^{-3}$Ns
23. $I = F \times t$
$= 200 \times 20$
$= 4000$
$= 4 \times 10^3$Ns
24. $I = \Delta P$
$I = m\Delta V$

$4000 = 200\Delta v$

$\Delta V = 20$

25. $I = \Delta P$

$I = 3.5 \times 10^5 \times 2$

$= 7 \times 10^5$ dyne $-$s or $= 7$Ns

26. $I = \Delta P$

$I = 3.5 \times 10^5 \times 2$

$\Delta P = 7 \times 10^5$ dyne $-$s

27. $I = m\Delta v$

$7 = 200 \times 10^{-3}\Delta v$

$7 = 0.2\Delta v$

$\Delta v = \frac{70}{2}$

$\Delta v = 35$ m/s

WORKSHEET – 6

Level – I

1. A	2. B	3. A	4. D	5. D	6. A
7. B	8. A	9. A	10. C		

Level – II

1. D	2. B	3. A	4. B	5. A	6. C
7. D	8. D	9. B	10. A	11. A	12. B
13. A	14. C	15. C	16. B		

Level – III

17. A	18. A	19. A	20. a-p; b-q; c-r; d-s		21. A
22. A	23. B	24. A	25. A	26. B	27. D
28. C					